

Report 14

Testing, Validation & Quality Assurance

Project:	JalJeev — Agentic Marine Intelligence System
Phase:	Continuous (Aug–Sep 2026)
Focus:	93-test pytest suite, risk hierarchy verification, vessel-class regression

1. Testing Philosophy

JalJeev is a safety-critical system — its recommendations directly influence whether a fisherman goes to sea. The testing strategy reflects this:

1. **Every hard constraint must be independently tested** — an MPA veto, a capsize-wave veto, a grounding veto
2. **The same ocean state must produce different correct answers for different vessels**
3. **The LLM-free fallback must pass every test the LLM path passes**
4. **No test should require network access or a running server** — all external data is mocked

2. Test Suite Overview

2.1 Location and Execution

```
cd backend && python -m pytest tests/ -v
```

- **93 tests** across multiple test files
- **Zero network dependencies** — all external calls mocked
- **Zero server dependency** — no running backend required
- **Execution time:** <15 seconds on local machine

2.2 Test File Inventory

File	Tests	Focus
test_risk_agent.py	~25	Hard constraint vetoes, scoring arithmetic, tier boundaries
test_vessel_classes.py	~20	Same conditions to different verdicts per vessel class
test_rule_based_planner.py	~15	Intent parsing, template rendering, edge cases
test_circuit_breaker.py	~10	Failure tracking, circuit open/close, recovery timeout
test_passage_agent.py	~12	Corridor geometry, H3 resolution selection
test_schemas.py	~5	Pydantic model validation, None vs zero
test_evidence.py	~6	Evidence receipt completeness, source tracking

3. Risk Hierarchy Tests

3.1 Hard Constraint (Stage 1) Tests

These verify that a veto **cannot** be overridden by favourable conditions elsewhere:

```
def test_mpa_veto_overrides_calm():
    result = risk_agent.assess_risk(...)
    assert result.risk_level == 'REJECTED'
    assert result.risk_score is None
```

Similar tests verify capsized-wave veto, grounding veto, and land veto independently and without network calls.

3.2 Scoring (Stage 2) Tests

Tests verify that calm conditions score LOW, multiple hazards stack correctly, boundary conditions (score 25 = MODERATE threshold) are handled, and factor breakdown is complete and non-negative.

4. Vessel Class Tests

The central insight: The same sea produces different correct verdicts for different vessels.

```
# 1.8m waves: dangerous for fishing_small, routine for bulk_carrier
fishing_result = risk_agent.assess_risk(ocean, ..., fishing_small)
assert fishing_result.risk_score > 40

builker_result = risk_agent.assess_risk(ocean, ..., bulk_carrier)
assert builker_result.risk_score < 15
```

Additional tests verify sail-specific factors (becalmed penalty applies to sailing vessels, not motors) and commercial vessel depth rules (squat calculation, under-keel clearance).

5. Rule-Based Planner Tests

test_rule_based_planner.py verifies the entire LLM-free fallback:

- **Intent parsing:** Regex-based intent detection for safety, PFZ, route, passage queries
- **Template rendering:** The answer only contains fields from EvidenceReceipt — no hallucinated data
- **Edge cases:** Missing sources, empty factor breakdown, None values handled correctly

6. Circuit Breaker Tests

test_circuit_breaker.py verifies that a slow or failing data source is temporarily disabled:

```
# Circuit opens after threshold
cb.record_failure('copernicus') # × 5
assert cb.is_open('copernicus') is True

# Success resets counter
cb.record_success('copernicus')
assert cb.is_open('copernicus') is False
```

7. End-to-End Validation

7.1 Live Verification Points

Scenario	Coordinates	Expected Result
Offshore Visakhapatnam	17.65, 83.35	Non-vetoed, real risk score
Bhubaneswar (inland)	20.30, 85.82	REJECTED (far from port)
Puri city centre	19.80, 85.83	REJECTED (on land)
Puri offshore	19.70, 85.83	Non-vetoed (26m depth)
Inside an MPA	(synthetic)	REJECTED (MPA veto)

7.2 Prewarm & Demo Verification

scripts/prewarm_demo.py doubles as a verification tool: queries every demo coordinate, verifies non-empty responses, confirms Redis cache population, reports adapter failures.

8. Validation Strategy for Judges

Validation Method	What It Proves
Live query with real coordinates	System fetches real data, not cached fakes
Bhubaneswar to REJECTED	System correctly refuses inland points
Vessel swap demo	Same location, different vessel gives different verdict
What-if scenario	Time-shift counterfactual shows reasoning
Kill the LLM key	System still works with rule-based fallback
Show evidence panel	Every number is traceable to a source

9. Coverage Assessment

Area	Coverage	Notes
Risk hierarchy (Stage 1 vetoes)	Good	Every veto type has tests
Risk scoring (Stage 2)	Good	Boundary conditions, stacking
Vessel classes	Comprehensive	11 classes, sail-specific factors
Rule-based fallback	Good	Intent parsing, rendering, edge cases
Circuit breaker	Good	Open/close/recovery
Passage geometry	Basic	Corridor, resolution, tacking
Adapter error handling	Partial	Happy paths and timeouts tested
Frontend	Not tested	No Vitest tests written yet

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